

Moduli and Deformation Theory

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1 Deformations and Stacks

With the Hilbert and Picard scheme, the definition of a first order deformation was what we expected, but for coherent sheaves the definition is perhaps slightly surprising—why do we have to track this data of an isomorphism?

We give a stacky justification.

1.1 Grothendieck Construction

Recall that there is a correspondence, called the *Grothendieck construction*, between

$$\text{sheaves valued in groupoids} \quad \longleftrightarrow \quad \text{stacks of groupoids}$$

where such a sheaf on a site $\mathcal{C}$

$$F : \mathcal{C}^{\text{op}} \rightarrow \text{Groupoid}$$

gets sent to the fibered category

$$\mathcal{X} \xrightarrow{\pi} \mathcal{C}$$

which has objects being pairs (C, X) where $C \in \mathcal{C}$ and $X \in F(C)$, and morphisms $(C, X) \rightarrow (C', X')$ being the data of a morphism $f : C \rightarrow C'$ in $\mathcal{C}$ inducing a functor

$$f^* : F(C') \rightarrow F(C)$$

together with an (iso)morphism $f^*X' \rightarrow X$. Finally, the functor π sends a pair (C, X) to just C .

You can then recover F by setting $F(C)$ to be the *fiber category*, i.e., objects being pairs (C, X) where C is our fixed object, but X can vary, and only allowing morphisms “over the identity”, that is, we force the map $f : C \rightarrow C$ to be the identity. You will then get pullback functors satisfying the necessary axioms by making a choice of pullback objectwise.

1.2 First Order Deformations as a Groupoid

From the groupoid perspective, it seems clear—a first order deformation of $x \in F(\text{Spec } k)$ should be an object $v \in F(\text{Spec } D)$ such that along the inclusion of the closed point $i :$

$\mathrm{Spec} k \rightarrow \mathrm{Spec} D$, we get

$$i^*v = x.$$

Notice the equality—this is the correct thing to do when F is a functor to Set . This still works for functors to $\mathrm{Groupoid}$, but in practice equality is very hard to work with—you need to make sure your algebro-geometric objects are nailed down on the nose.

Instead, recall that i^* is a functor of groupoids

$$F(\mathrm{Spec} D) \xrightarrow{i^*} F(\mathrm{Spec} k),$$

and we can then ask about the preimage of x , which should itself be a groupoid. Recall that preimages in algebraic geometry are constructed as fiber products, and so similarly we should define the preimage $(i^*)^{-1}(x)$ to be the following fiber product.

$$\begin{array}{ccc} (i^*)^{-1}(x) & \longrightarrow & F(\mathrm{Spec} D) \\ \downarrow & & \downarrow i^* \\ \mathbb{1} & \xrightarrow{x} & F(\mathrm{Spec} k) \end{array}$$

But we should review fiber products of groupoids.

1.3 Fiber Products of Groupoids

Suppose we have a diagram of groupoids

$$\begin{array}{ccc} & & \mathcal{U} \\ & & \downarrow f \\ \mathcal{V} & \xrightarrow{g} & \mathcal{W} \end{array}$$

You might imagine $\mathcal{U}, \mathcal{V}$ are categories of sheaves on open sets U, V , and $W = U \cap V$. The fiber product will be one way of constructing the category of sheaves on the union.

The fiber product $\mathcal{U} \times_{\mathcal{W}} \mathcal{V}$ should be characterized by a 2-universal property, which defines it only up to equivalence (not isomorphism). You can in particular get this 2-universal property by defining the fiber product in an “evil” way, where you take

- objects to be pairs $(u, v) \in \mathcal{U} \times \mathcal{V}$ such that $F(u) = G(v)$ with literal equality,
- morphisms $(u, v) \rightarrow (u', v')$ to be pairs of arrows $\varphi : u \rightarrow u', \psi : v \rightarrow v'$ such that $f(\varphi) = g(\psi)$.

but this is exactly the sort of set theory problem we want to avoid. For example, consider $\mathbb{P}^1_{[x:y]}$ with its cover $U = D_+(x) = \mathrm{Spec} k[\frac{y}{x}]$ and $V = D_+(y) = \mathrm{Spec} k[\frac{x}{y}]$. To specify a nontrivial line bundle like $\mathcal{O}(1)$, it is *not* enough to write down isomorphism classes of the line bundle restricted to opens such that they happen to be isomorphic on the overlap—that is, the data of $\mathcal{O}_U$ and $\mathcal{O}_V$ could have actually been the local data for any line bundle $\mathcal{O}(d)$.

Instead, our “evil” approach rectifies this lack of data by saying we cannot just remember an isomorphism class—we need to honestly remember sheaves on a literal opens U, V (not just isomorphic schemes) of $\mathbb{P}^1$. Then, with $\mathcal{O}(1)|_U, \mathcal{O}(1)|_V$ we can actually recover $\mathcal{O}(1)$, since to define a sheaf you only need to know its sections/restriction maps over open sets forming a base of the topology, and open subsets of U, V form a base.

This is confusing—the much more familiar approach to defining a line bundle on an open cover is to define transition functions to rigidify our local data. This motivates this non-“evil” definition of the fiber product of groupoids. Define $\mathcal{U} \times_{\mathcal{V}} \mathcal{V}$ to be the category with

- objects to be triples (u, v, α) where $u \in \mathcal{U}, v \in \mathcal{V}$, and $\alpha : f(u) \rightarrow g(v)$ a chosen isomorphism.
- morphisms $(u, v, \alpha) \rightarrow (u', v', \alpha')$ to be pairs (φ, ψ) where $\varphi : f(u) \rightarrow f(u'), \psi : g(v) \rightarrow g(v')$ such that the following diagram commutes.

$$\begin{array}{ccc} f(u) & \xrightarrow{f(\varphi)} & f(u') \\ \downarrow \alpha & & \downarrow \alpha' \\ g(v) & \xrightarrow{g(\psi)} & g(v') \end{array}$$

1.4 Non-Evil First Order Deformations

Now let’s apply our knowledge. Repeating the definition in our special case, the first order deformations are the groupoid, which is the preimage $(i^*)^{-1}(x)$. Denoting the unique object $\bullet \in \mathbb{1}$, our definition of the fiber product $F(\mathrm{Spec} D) \times_{F(\mathrm{Spec} k)} \mathbb{1}$ specializes to the category with

- objects $(v, \bullet, \alpha)$, where $v \in F(\mathrm{Spec} D)$ and $\alpha : i^*(v) \rightarrow x$ an isomorphism
- morphisms $(v, \bullet, \alpha) \rightarrow (v', \bullet, \alpha')$ to be pairs $(\varphi, 1_\bullet)$ where $\varphi : i^*(v) \rightarrow i^*(v')$ such that the following diagram commutes.

$$\begin{array}{ccc} i^*(v) & \xrightarrow{i^*(\varphi)} & i^*(v') \\ \downarrow \alpha & & \downarrow \alpha' \\ x & \xrightarrow{1_x} & x \end{array}$$

We might as well omit the bullet $\bullet$ and just write our objects as pairs (v, α) .

For example, we can consider the stack/sheaf of groupoids of coherent sheaves on X (which is, say, a proper Noetherian k -scheme to make sure our definitions are correct). Denote this gadget by Coh_X . This will have the fiber $\mathrm{Coh}_X(T)$ over a k -scheme T to be the groupoid of coherent sheaves on $X \times_k T$ flat over T . Given a morphism $T' \xrightarrow{f} T$, we get a pullback functor f^* from coherent sheaves on $X \times T$ to those on $X \times T'$.

In this setup, the groupoid of first order deformations of $\mathcal{F} \in \mathrm{Coh}_X(\mathrm{Spec} k)$ will be a pair $(\mathcal{F}', \alpha)$, where $\mathcal{F}' \in \mathrm{Coh}_X(\mathrm{Spec} D)$ is a coherent sheaf on $X \times_k \mathrm{Spec} D$ flat over $\mathrm{Spec} D$, and

for $i : \operatorname{Spec} k \rightarrow \operatorname{Spec} D$ the inclusion of the closed point, α is an isomorphism

$$i^* \mathcal{F}' \xrightarrow{\alpha} \mathcal{F}.$$

Finally, a morphism of first order deformations $(\mathcal{F}'_1, \alpha_1) \rightarrow (\mathcal{F}'_2, \alpha_2)$ is a morphism $\mathcal{F}'_1 \xrightarrow{\varphi} \mathcal{F}'_2$ such that the triangle commutes.

$$\begin{array}{ccc} i^* \mathcal{F}_1 & \xrightarrow{\varphi} & i^* \mathcal{F}_2 \\ \searrow \alpha_1 & & \swarrow \alpha_2 \\ & \mathcal{F} & \end{array}$$

This exactly recovers Hartshorne's definition of a first order deformation of a coherent sheaf, and our task is to classify these objects up to isomorphism (i.e., find the connected components of this groupoid).

2 Hypersurfaces

Fix a field k and a pair of dual $n + 1$ -dimensional vector spaces V, W over k . I will sometimes like to think of V as “geometric” and W as “coordinates”, but sometimes it will be convenient to swap this view.

Our task is to study the moduli space of degree d hypersurfaces of $\mathbb{P}V$ (the projective space of lines in V).

This is one example of a Hilbert scheme, where your Hilbert polynomial is

$$P(m) = h_{\mathcal{O}_{\mathbb{P}V}}(m) - h_{\mathcal{O}_{\mathbb{P}V}(-d)}(m) = \binom{n+m}{n} - \binom{n+m-d}{n}.$$

Proposition 2.1. $\mathbb{P}\operatorname{Sym}^d W$ is the Hilbert scheme of degree d hypersurfaces.

We give a discussion to motivate this result.

2.1 Intuition

Concretely, we should think of $\mathbb{P}V$ as our usual $\mathbb{P}^n$, where $\mathbb{P}^n$ in turn is Proj of the polynomial ring $\operatorname{Sym}^\bullet W = k[x_0, \dots, x_n]$ (notice the dual).

We are then claiming the moduli space of degree d hypersurfaces is

$$\mathbb{P}k[x_0, \dots, x_n]_d$$

the projective space of lines in the vector space $k[x_0, \dots, x_n]_d$. We should guess this because a k -point of $\mathbb{P}\operatorname{Sym}^d W$ is a line through $\operatorname{Sym}^d W$, i.e., the span of a degree d polynomial $f \in k[x_0, \dots, x_n]_d$.

2.2 Projective space as a moduli space

Most classically, we like to think of the projective space $\mathbb{P}V$ as the set of 1-dimensional subspace of V —this already sounds like a moduli space, since $\mathbb{P}V$ is parameterizing some algebro-geometric object. However, we should see that this respects families in some sense.

Complicating things, it will turn out to be slightly better to think $\mathbb{P}V$ actually as the set of 1-dimensional quotients of the dual W . This is really equivalent, since for L 1-dimensional, $L \hookrightarrow V$ is the same data as $W \twoheadrightarrow L^\vee$.

The reason for the switch is essentially because $(-)|_p \cong (-) \otimes_{\mathcal{O}_T} k(p)$ preserves surjections, not injections (but this is only because we allow “more” morphisms of sheaves than we classically allow for vector bundles a la differential geometry).

To formulate $\mathbb{P}V$ as the moduli space of 1-dimensional quotients of W , we first remember how to write maps to projective space.

Remark 2.2. Recall that a morphism to projective space $T \rightarrow \mathbb{P}k^{n+1}$ is exactly the same data as a line bundle L on T together with a surjection (global generation of)

$$\mathcal{O}_T \otimes_k k^{n+1} \twoheadrightarrow L.$$

Informally, this works because at each point $t \in T$, the surjection gives values to $x_0, \dots, x_n$ in $L|_t \cong k$ (where at least one $x_i \neq 0$ because of surjectivity), which gives a point $[x_0 : \dots : x_n]$ in projective space. It is no problem that $L|_t$ is not literally k and needs an identification, since we only needed to find the point up to scaling anyways.

This L will then be the pullback of $\mathcal{O}(1)$, which is the “universal” way of assigning values of $x_0, \dots, x_n$ in projective space $\mathbb{P}^n$.

More formally, we get a map to projective space as follows. The surjection $\mathcal{O}_T \otimes_k k^{n+1} \rightarrow L$ is a row matrix

$$[f_0 \ f_1 \ \cdots \ f_n]$$

where each f_i is a global section of L , and surjectivity means that there are no common zeros among the f_i . Essentially, we want to define

$$\begin{aligned} T &\rightarrow \mathbb{P}k^{n+1} \\ t &\mapsto [f_0(t) : f_1(t) : \cdots : f_n(t)], \end{aligned}$$

where all of the $f_i(t) \in L|_t$, which we can identify with k in any way we want (since all identifications are the same up to scaling). However, we really want to define this scheme theoretically.

So, given the data of L with this surjection, we produce a map $T \rightarrow \mathbb{P}k^{n+1}$ by first noting that on $D(f_i)$, L becomes trivialized because

$$\mathcal{O}_{D(f_i)} \xrightarrow{f_i} L|_{D(f_i)}$$

is an isomorphism. Then, on $D(f_i)$, we would want to be able to write down something like

$$D(f_i) \rightarrow \mathbb{P}k^{n+1}$$

$$t \mapsto \left[\frac{f_0(t)}{f_i(t)} : \cdots : \frac{f_n(t)}{f_i(t)} \right] \in D_+(x_i) \cong \mathbb{A}^n,$$

and this time we really can make sense of this scheme theoretically, by defining the map $D(f_i) \rightarrow \mathbb{A}^n$. Recall that a map to affine n -space is the same data as giving n -functions, so we just write down

$$\frac{f_0}{f_i}, \dots, \frac{f_n}{f_i}$$

(omitting $\frac{f_i}{f_i} = 1$). Then, one should check that these maps $D(f_i) \rightarrow U_i$ assemble into a map $T \rightarrow \mathbb{P}k^{n+1}$ by checking the pieces identify on restriction.

Remark 2.3. We can rephrase the above in a coordinate-free way. A morphism

$$T \rightarrow \mathbb{P}V$$

is (still) equivalent to the data of a line bundle L on T and a surjection

$$\mathcal{O}_T \otimes_k W \xrightarrow{e} L.$$

One can show that this works by picking coordinates, but it is useful to note that the resulting set function on k -points can be described more intrinsically as

$$T \rightarrow \mathbb{P}V$$

$$t \mapsto \langle e|_p : W \rightarrow L|_p \rangle,$$

where $e|_p$ is a 1-dimensional quotient of W , which as discussed is the same data as a 1-dimensional subspace of V .

Finally, slightly reformulating the previous discussion, we are just saying that $\mathbb{P}V$ is the moduli space of 1-dimensional quotients of W . An individual 1-dimensional quotient is a surjection

$$W \rightarrow L,$$

where L is a vector space, or equivalently a line bundle on $\text{Spec } k$.

Correspondingly, a family over T of 1-dimensional quotients is a surjection

$$\mathcal{O}_T \otimes W \rightarrow L,$$

where now L is a line bundle on T . This becomes a moduli functor $F : (\text{Sch}/k)^{\text{op}} \rightarrow \text{Set}$ sending T/k to all such surjections.

Our previous discussion gives a bijection between

$$\text{Maps } T \rightarrow \mathbb{P}V \quad \text{surjections } \mathcal{O}_T \otimes W \rightarrow L,$$

which is in fact natural (commutes with pullbacks $S \rightarrow T$), so this gives a natural isomorphism between the moduli functor F and $\text{Hom}(-, \mathbb{P}V)$, so indeed $\mathbb{P}V$ is the moduli space of 1-dimensional quotients of W .

2.3 Proof (in many words)

Now we show that $\mathbb{P}\mathrm{Sym}^d W$ is the Hilbert scheme of degree d hypersurfaces. Thinking moduli-theoretically, we want to establish a bijection between the Hilbert scheme functor and the moduli functor of 1-dimensional quotients of $(\mathrm{Sym}^d W)^\vee$.

That is, given a family of hypersurfaces over a k -scheme T

$$\begin{array}{ccc} Y & \subseteq & \mathbb{P}V \times T \\ & \searrow & \downarrow \\ & & T \end{array}$$

($Y \rightarrow T$ flat), we should cook up a corresponding surjection

$$\mathcal{O}_T \otimes_k (\mathrm{Sym}^d W)^\vee \xrightarrow{e} L.$$

We should then check that this procedure is a bijection and that it is *natural*, i.e., commute with pullbacks along a morphism $S \rightarrow T$.

2.3.1 Along a fiber

At the end of the day, this procedure is just copying the “intuition”, but instead in families. So, we start by thinking carefully about the case when $T = \mathrm{Spec} k$ so that a family of hypersurfaces is just one hypersurface. Let $Y \subseteq \mathbb{P}V$ be a $\mathrm{Spec} k$ point of the Hilbert scheme. It will have a defining degree d equation f , and we wanted to send Y to

$$\langle f \rangle \hookrightarrow k[x_0, \dots, x_n]_d \cong \mathrm{Sym}^d W,$$

or dually

$$(\mathrm{Sym}^d W)^\vee \twoheadrightarrow \langle f \rangle^\vee.$$

In aims of generalizing, we should recognize that the defining equation f comes as a section

$$f \in \Gamma(\mathbb{P}V, \mathcal{I}_Y(d)) \subseteq \Gamma(\mathbb{P}V, \mathcal{O}_{\mathbb{P}V}(d)) = \mathrm{Sym}^d W,$$

so the inclusion of the 1-dimensional subspace is

$$\Gamma(\mathbb{P}V, \mathcal{I}_Y(d)) \hookrightarrow \mathcal{O}_{\mathbb{P}V}(d),$$

and the 1-dimensional quotient is just the dual. To do this more generally, we will just realize that Γ is pushforward to our base $T = \mathrm{Spec} k$

2.3.2 In total

We try to repeat the strategy from before, but do it over all of T rather than just over an individual k -point $t \in T$. We first fix notation for projections.

$$\begin{array}{ccc}
& \mathbb{P}V \times T & \\
p \swarrow & & \searrow q \\
\mathbb{P}V & & T
\end{array}$$

Now, to repeat the strategy, we should take the short exact sequence cutting out $Y \subseteq \mathbb{P}V \times T$

$$0 \rightarrow \mathcal{I}_Y \rightarrow \mathcal{O}_{\mathbb{P}V \times T} \rightarrow \mathcal{O}_Y \rightarrow 0,$$

then twist by d to study the degree d part

$$0 \rightarrow \mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d) \rightarrow p^* \mathcal{O}(d)_{\mathbb{P}V} \rightarrow \mathcal{O}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d) \rightarrow 0,$$

then pushforward to T to turn things into vector spaces (smeared along T , i.e., a sheaf) to get an injection

$$q_*(\mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d)) \hookrightarrow q_*(p^* \mathcal{O}_{\mathbb{P}V}(d)).$$

Flat base change will tell you

$$q_*(p^* \mathcal{O}_{\mathbb{P}V}(d)) \cong \mathcal{O}_T \otimes_k \text{Sym}^d W,$$

and some playing with cohomology and base change will show $q_*(\mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d))$ is a line bundle, but we postpone the details until later.

Finally, we should take duals with $\mathcal{H}om(-, \mathcal{O}_T)$ and show that this is a surjection to get our family of 1-dimensional quotients of $(\text{Sym}^d W)^\vee$.

2.3.3 Applying Cohomology and Base Change

Application 1

Now, we show that $q_*(\mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d))$ is a line bundle. First, $\mathcal{I}_Y = \mathcal{O}_{\mathbb{P}V \times T}(-Y)$ is a line bundle because it is the ideal sheaf of a Cartier divisor. So,

$$\mathcal{O}_{\mathbb{P}V \times T}(-Y) \otimes p^* \mathcal{O}_{\mathbb{P}V}(d) \in \text{Coh}(\mathbb{P}V \times T)$$

is flat over T because it is a line bundle on $\mathbb{P}V \times T$, which is itself flat over T .

Now, cohomology and base change theorems [1, Theorem 25.1.6] will tell us that over every point $t \in T$ (or just closed points, or dense points), if we can prove the “restriction” morphism

$$q_*(\mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d))|_t \rightarrow H^0(\mathbb{P}V \times \text{Spec } k(t), (\mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d))|_{\mathbb{P}V \times \text{Spec } k(t)})$$

is surjective, then the pushforward is a vector bundle with rank the same as either vector space (which will have to be isomorphic). This would be applying the base change theorem in degree 0. To get this surjectivity, we in turn apply the base change theorem in higher degree, so we need surjectivity from a $R^\ell q_*(-)|_t$ to an H^ℓ .

Now, recognize that the right hand side is the cohomology of

$$(\mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d))|_{\mathbb{P}V \times \text{Spec } k(t)} \cong \mathcal{O}_{\mathbb{P}V \times \text{Spec } k(t)},$$

essentially the Y and d cancel out in the fiber. More formally, we are taking the twisted ideal sheaf short exact sequence

$$0 \rightarrow \mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d) \rightarrow p^* \mathcal{O}_{\mathbb{P}V}(d) \rightarrow \mathcal{O}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d) \rightarrow 0$$

and restricting to $\mathbb{P}V \times \text{Spec } k(t)$ (i.e., tensoring over $\mathcal{O}_{\mathbb{P}V \times T}$ with $\mathcal{O}_{\mathbb{P}V \times \text{Spec } k(t)}$), which will still be exact because the flatness of $\mathcal{O}_Y$ will make a $\mathcal{T}or_1$ vanish (**I think my justification is a little bogus here, I need to figure this out**), so the restricted twisted ideal is the kernel of

$$\mathcal{O}_{\mathbb{P}V \times \text{Spec } k(t)}(d) \rightarrow \mathcal{O}_{Y_t}(d),$$

which is $\mathcal{O}_{\mathbb{P}V \times \text{Spec } k(t)}$, since Y_t is a degree d hypersurface.

Now, $\mathcal{O}_{\mathbb{P}V \times \text{Spec } k(t)}$ has dimension 1 cohomology in degree 0, but no higher cohomology. Now we apply cohomology and base change in degree 1—the vanishing for this right hand side sheaf fiber will force the restriction to be surjective/an isomorphism, forcing the higher direct image to be 0 and in particular locally free, which forces our surjection in degree 0.

Thus, $q_*(\mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d))$ is locally free of rank 1.

Application 2

Now, we show that the dual to the map

$$q_*(\mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d)) \xleftarrow{i} q_*(p^* \mathcal{O}_{\mathbb{P}V}(d))$$

is surjective. The obstruction to surjectivity is $\mathcal{E}xt_{\mathcal{O}_T}^1(\text{coker } i, \mathcal{O}_T)$, and we would hope $\text{coker } i$ would just be q_* applied to the cokernel upstairs, i.e.,

$$q_*(\mathcal{O}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d)),$$

and the obstruction to this is the term

$$R^1 q_*(\mathcal{I}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d)).$$

We will succeed on both accounts. First, to get this $R^1 q_*$ term to vanish, argue by cohomology and base change literally as before (we had already analyzed that the relevant H^1 s vanish, so that the fibers of $R^1 q_*$ vanish, so $R^1 q_*(-) = 0$).

Next,, we get the $\mathcal{E}xt^1$ term to vanish by arguing

$$\text{coker } i = q_*(\mathcal{O}_Y \otimes p^* \mathcal{O}_{\mathbb{P}V}(d))$$

is locally free, again by cohomology and base change, getting a surjectivity of restriction to fibers in degree 0, which follows from $R^1 q_*$ vanishing because it has 0 fibers.

References

- [1] Ravi Vakil. *The rising sea: foundations of algebraic geometry*. Princeton: Princeton University Press, 2025. 1 p. ISBN: 978-0-691-26867-5 978-0-691-26868-2.